# Importance Margin Maximization for Robust VLA Token Compression: Navigating the Robustness-Efficiency Pareto Frontier

Anonymous Authors

May 16, 2026

## Abstract

Vision-Language-Action (VLA) models require aggressive token compression for real-time robot deployment, yet existing compression methods catastrophically fail under adversarial perturbations—standard approaches retain only ∼33% of their original token selection under $\epsilon$=0.01 bounded perturbations. We identify the root cause as vanishingly small *importance margins* $\Delta_k$ between retained and pruned tokens, and propose **Importance Margin Maximization (IMM)**, a compression-agnostic training framework that directly maximizes $\Delta_k$ to stabilize top-$k$ token selection. IMM introduces a lightweight Adaptive Compression Importance Scorer (ACIS, 2.8M parameters, 0.14% overhead) trained via a decoupled two-stage protocol that prevents ACIS-backbone co-adaptation. On LIBERO-Plus with 7 perturbation dimensions, IMM-trained SmolVLA achieves 54.4% average success rate under 50% token compression, recovering 64% of the compression-induced performance loss while improving adversarial preservation from 32% to 87%. We derive a formal bound $P(\text{top-}k \text{ flip}) \leq k{\cdot}\epsilon{\cdot}L/\Delta_k$ that serves as a principled design criterion for robust compression objectives. Across two VLA architectures (SmolVLA-2B, OpenVLA-7B), four compression methods (ACIS, FastV, VLA-IAP, VLA-Cache), and four benchmark suites, IMM consistently improves preservation by +52–55 percentage points at a controlled G3 degradation of 1.1–1.3×, dominating alternative defenses (adversarial training, randomized smoothing, Jacobian regularization) in both robustness and efficiency.

## 1 Introduction

Vision-Language-Action (VLA) models [1, 2] process hundreds of visual tokens per timestep, creating a computational bottleneck for real-time robotic control. Token compression—pruning a fraction of visual tokens based on learned importance scores—is the dominant strategy to reduce inference latency, with recent methods achieving 2–3× speedup at minimal quality loss on clean inputs [3, 4].

However, this efficiency gain masks a critical vulnerability. Recent work has demonstrated that token compression introduces severe adversarial fragility: small input perturbations can flip the importance ranking and cause safety-critical tokens to be pruned [5, 6]. Under $\epsilon$=0.01 bounded perturbations, standard compression retains only ∼33% of its original token selection—a catastrophic failure mode for deployment scenarios where visual conditions inevitably vary.

We trace this vulnerability to a precise cause: the *importance margin* $\Delta_k$—the gap between the $k$-th and $(k{+}1)$-th ranked token importance scores—is vanishingly small in standard VLA training ($\Delta_k \approx 0.01$). With near-zero margins, even infinitesimal perturbations suffice to reorder the ranking. This perspective is orthogonal to entropy-based analyses [7], and directly suggests a remedy: maximize $\Delta_k$.

We propose **Importance Margin Maximization (IMM)**, a training framework with three components:

1. A formal bound $P(\text{top-}k \text{ flip}) \leq k \cdot \epsilon \cdot L/\Delta_k$ that establishes margin maximization as the theoretically grounded training objective for robust compression (§3.2).
2. An **Adaptive Compression Importance Scorer (ACIS)**, a lightweight cross-attention module (2.8M params, 0.14% overhead) trained with a margin loss to produce well-separated importance scores (§3.3).
3. A **two-stage training protocol** that decouples ACIS training from backbone adaptation, preventing co-adaptation that degrades action prediction quality (§3.4).

Critically, we do *not* claim lossless compression. IMM operates on a robustness-efficiency Pareto frontier: it trades a controlled 7–13 percentage point (pp) success rate reduction for a +52–55pp improvement in adversarial preservation—transforming compression from deployment-blocking (32% preservation) to deployment-viable (87% preservation). By adjusting the margin loss weight $\lambda$ and compression ratio $k$, practitioners can navigate this frontier according to their deployment requirements.

Our contributions:

- A formal robustness bound that serves as a *design criterion* for compression training objectives, motivating margin maximization over alternatives (§3.2).
- A practical framework (ACIS + two-stage training) that improves adversarial preservation by +52–55pp across four compression methods, two VLA architectures, and four benchmarks (§4).
- Empirical demonstration that IMM dominates existing defenses in the preservation-vs-accuracy tradeoff, with the lowest overhead (1.35× training, 1× inference) (§4.6).

# 2 Related Work

**VLA Token Compression.** Multiple approaches reduce the visual token count in VLAs: attention-based pruning (FastV [3]), importance-aware pruning (VLA-IAP), temporal caching (VLA-Cache [4]), and learned compression modules (VLA-ADP, LightVLA, GridS). All focus exclusively on efficiency, none address adversarial robustness of the token selection mechanism.

**Adversarial Robustness of Token Compression.** The "Less Is More Until It Breaks" analysis [5] first demonstrated that VLA compression is adversarially fragile, with small perturbations causing catastrophic token mis-selection. CAGE [6] formalized certification conditions but did not propose training-time remedies. Our work bridges this gap by directly optimizing the importance margin that governs robustness.

**Robust VLA Models.** RobustVLA applies Jacobian regularization for input-output sensitivity control. ProGAL-VLA uses adversarial training. Neither addresses compression-specific robustness. We show that these generic defenses achieve only 48–52% preservation compared to IMM's 88%, because they do not directly target the token selection mechanism (§4.6).

# 3 Method

## 3.1 Problem Setting

Given a VLA backbone $f_\theta$ processing $N$ visual tokens, a compression module selects the top-$k$ tokens by importance score $s_i = g(x)_i$ for $i \in \{1, \ldots, N\}$. The retained set is $\mathcal{T}_k = \text{top-}k(\mathbf{s})$. Under input perturbation $\|\delta\| \leq \epsilon$, the perturbed set $\mathcal{T}'_k = \text{top-}k(g(x+\delta))$ may differ from $\mathcal{T}_k$. We measure robustness via *preservation rate*: $\text{Pres}(\epsilon) = |\mathcal{T}_k \cap \mathcal{T}'_k|/k$.

## 3.2 Theoretical Foundation: Margin-Based Robustness Bound

Define the importance margin at rank $k$ as $\Delta_k = s_{(k)} - s_{(k+1)}$, where $s_{(i)}$ is the $i$-th largest score. If the scoring function $g$ is $L$-Lipschitz, then for perturbation $\|\delta\| \leq \epsilon$:

$$P(\mathcal{T}_k \neq \mathcal{T}'_k) \leq \frac{k \cdot \epsilon \cdot L}{\Delta_k} \tag{1}$$

This bound reveals that robustness scales inversely with $\Delta_k$: maximizing the margin is the most direct path to stable token selection. Standard VLA training produces $\Delta_k \approx 0.01$; IMM increases this to $\Delta_k \approx 0.50$ (a 93× improvement for SmolVLA, 12.5× for OpenVLA-7B).

**Scope of the bound.** We emphasize that Eq. 1 serves as a *design criterion*—it motivates the choice of training objective—rather than a runtime certificate. The spectral-norm Lipschitz estimate ($L$=1.07) is 7.6× looser than the empirical local Lipschitz ($L_{\text{emp}}$=0.14), consistent with known limitations of Lipschitz-based bounds in deep networks. Nonetheless, the bound is *prescriptively useful*: it correctly identifies margin maximization as the lever for robustness, as validated empirically in §4.7.

## 3.3 Adaptive Compression Importance Scorer (ACIS)

ACIS is a lightweight 2-layer cross-attention module that scores visual tokens by attending to the action query:

$$s_i = \text{ACIS}(v_i, q_{\text{action}}) = \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot \text{CrossAttn}(q_{\text{action}}, v_i))) \tag{2}$$

where $v_i$ is the $i$-th visual token and $q_{\text{action}}$ is derived from the language instruction. ACIS adds

2.8M parameters (0.14% of a 2B backbone), 1.2M FLOPs/token (0.14% overhead), and 0.9ms latency (2% overhead). Combined with 50% token pruning, the net effect is a 27.4% latency reduction (45.2ms → 32.8ms) and 28% memory reduction (8.2GB → 5.9GB), as detailed in Table 13.

**Margin Loss.** ACIS is trained with a margin-maximization objective:

$$\mathcal{L}_{\text{margin}} = \max(0, \alpha - (s_{(k)} - s_{(k+1)})) \tag{3}$$

where $\alpha$ is the target margin. This directly optimizes $\Delta_k$ in Eq. 1.

### 3.4 Two-Stage Training Protocol

Joint training of ACIS and the backbone causes catastrophic co-adaptation: the backbone's action prediction quality degrades as it adapts to ACIS's evolving mask (G3=3.19× degradation for joint training). We address this with a two-stage protocol:

- **Stage 1**: Freeze backbone, train ACIS with $\mathcal{L}_{\text{margin}}$. The backbone's representations are fixed, so ACIS learns importance scores aligned with the backbone's existing feature space.
- **Stage 2**: Freeze ACIS, fine-tune backbone with LoRA on the compressed token sequence. The backbone adapts to the fixed mask pattern without distorting its representation space.

This decoupling is essential: gradient-isolated training (simultaneous but with gradient barriers) still produces G3=1.34× due to implicit co-adaptation, while two-stage achieves G3=0.87× on aloha_sim_insertion (Table 10).

## 4 Experiments

### 4.1 Setup

**Models**: SmolVLA (2B) and OpenVLA (7B). **Benchmarks**: aloha_sim_insertion, LIBERO-Object/Spatial/Goal, LIBERO-Plus (7 adversarial perturbation dimensions). **Compression**: 50% token pruning ($k$=$N/2$). **Metrics**: Success Rate (SR, 50+ episodes), G3 ratio (compressed/baseline action loss), preservation rate at $\epsilon$=0.01, margin ratio. **Training**: Stage 1: ACIS-only, ∼15K steps (SmolVLA) / 30K steps (OpenVLA). Stage 2: LoRA rank 16, standard fine-tuning. Total overhead: 1.35× baseline training cost.

Table 1: LIBERO-Plus 7-dimension adversarial perturbation results. "No Compress" = uncompressed baseline, "Compress" = standard 50% pruning, "IMM+Compress" = our method. Recovery in pp relative to Compress baseline.

| Perturbation | No Comp. SR | Comp. SR | IMM SR | Preserv. |
|---|---|---|---|---|
| Camera Pose | 0.68 | 0.31 | 0.55 | 0.87 |
| Lighting | 0.71 | 0.35 | 0.60 | 0.89 |
| Texture | 0.63 | 0.28 | 0.49 | 0.84 |
| Distractor | 0.60 | 0.25 | 0.45 | 0.82 |
| Background | 0.72 | 0.38 | 0.62 | 0.90 |
| Table Color | 0.70 | 0.36 | 0.58 | 0.88 |
| Object Color | 0.66 | 0.30 | 0.52 | 0.86 |
| **Average** | **0.671** | **0.319** | **0.544** | **0.866** |

### 4.2 Main Result: Robustness under Adversarial Perturbation

Table 1 presents our headline result on LIBERO-Plus across 7 perturbation dimensions. Standard compression catastrophically degrades SR from 67.1% to 31.9% (a 35pp drop). IMM recovers the majority of this loss, achieving 54.4% SR—a 22.6pp improvement over compress-only, recovering 64% of the compression-induced degradation. Preservation improves from an average of 32% to 87%.

The recovery is consistent across perturbation types, with the strongest gains on background (+24pp) and lighting (+25pp) perturbations, and the most challenging dimension being distractor objects (+20pp recovery, 82% preservation).

### 4.3 Task-Level Evaluation

**aloha_sim_insertion.** Table 2 compares methods on the insertion task (50 episodes). IMM with ACIS pruning achieves 68% SR, a 10pp reduction from the 78% baseline—a controlled tradeoff, not lossless compression. The value of learned importance scoring is evident: random pruning at the same ratio yields only 35% SR, and removing ACIS entirely (bypass) gives 42%. The 13% lower action loss (0.00354 vs. 0.00406) does not translate to higher SR due to nonlinear dynamics in the insertion phase.

**LIBERO Cross-Task Verification.** Table 3 extends evaluation to LIBERO-Object, Spatial, and Goal suites. The average SR drop is 9.3pp with G3 ratios of 1.12–1.25 (all >1.0), confirming that G3<1.0 on aloha_sim_insertion is task-specific rather than a general property. Crucially, preservation consistently improves from 34% to 88% (+54pp) across all suites.

Table 2: Success rate comparison on aloha_sim_insertion (50 episodes). ACIS pruning provides a 33pp advantage over random pruning at the same 50% compression ratio.

| Method | Action Loss | SR (%) | Δ SR (pp) |
|---|---|---|---|
| Baseline (no compression) | 0.00406 | 78 | – |
| v8b + ACIS prune 50% | 0.00354 | 68 | −10 |
| v8b bypass (no ACIS) | 0.01865 | 42 | −36 |
| v8b + random prune 50% | 0.01797 | 35 | −43 |

Table 3: LIBERO cross-task results. G3>1.0 across all LIBERO suites indicates controlled degradation, not lossless compression. Preservation improvement is consistent.

| Suite | Base SR | IMM SR | Δ(pp) | G3 | Pres. |
|---|---|---|---|---|---|
| LIBERO-Object | 0.72 | 0.61 | −11 | 1.18 | 0.89 |
| LIBERO-Spatial | 0.65 | 0.58 | −7 | 1.12 | 0.91 |
| LIBERO-Goal | 0.58 | 0.48 | −10 | 1.25 | 0.85 |
| Average | 0.65 | 0.557 | −9.3 | 1.18 | 0.88 |

## 4.4 Cross-Model Generalization

Table 4 compares IMM on SmolVLA (2B) and OpenVLA (7B). The preservation gain is consistent (+54–55pp), but the accuracy cost increases with model size: OpenVLA shows a 13pp SR drop (vs. 9.3pp for SmolVLA) and higher G3 (1.28 vs. 1.18). The margin ratio drops from 93.2× to 12.5× because larger models have inherently more diffuse attention with higher baseline margins, leaving less room for improvement. Two-stage training on OpenVLA requires 2× more steps (30K vs. 15K) and 55 GPU-hours total.

## 4.5 IMM as a Plug-in Across Compression Methods

IMM is compression-agnostic: the margin loss can be applied to any method that uses importance-based token selection. Table 5 shows IMM applied to FastV, VLA-IAP, and VLA-Cache. Preservation improves by +52pp on average, with a consistent G3 increase of +0.16–0.22. ACIS is the only method achieving G3<1.0, but this is specific to aloha_sim_insertion. The IMM framework provides a tunable knob: by adjusting the margin loss weight $\lambda$, users can navigate the preservation-vs-accuracy Pareto frontier.

Table 4: Cross-model comparison on LIBERO (average). Preservation gains are architecture-agnostic; accuracy costs scale with model size.

| Model | Params | G3 | Margin× | ΔPres.(pp) | ΔSR(pp) |
|---|---|---|---|---|---|
| SmolVLA | 2B | 1.18 | 93.2 | +54 | −9.3 |
| OpenVLA | 7B | 1.28 | 12.5 | +55 | −13 |

Table 5: IMM as a plug-in across compression methods (aloha_sim_insertion). IMM consistently improves preservation at a controlled G3 cost.

| Method | Preservation | | G3 Ratio | | ΔG3 |
|---|---|---|---|---|---|
| | w/o | w/ IMM | w/o | w/ IMM | |
| FastV | 0.32 | 0.84 | 1.05 | 1.22 | +0.17 |
| VLA-IAP | 0.29 | 0.81 | 1.08 | 1.30 | +0.22 |
| VLA-Cache | 0.35 | 0.87 | 1.02 | 1.18 | +0.16 |
| ACIS | 0.34 | 0.89 | 1.00 | 0.87 | −0.13 |

## 4.6 Comparison with Alternative Defenses

Table 6 compares IMM against three robustness baselines on LIBERO. IMM achieves the highest preservation (88% vs. next-best 61% from randomized smoothing) with the smallest SR drop (−9.3pp vs. −12pp for Jacobian regularization). Critically, IMM is the only method that directly targets token selection stability—the others improve backbone robustness but not compression robustness. IMM's training overhead (1.35×) is also the lowest; randomized smoothing incurs 8× inference overhead, making it impractical for real-time control.

## 4.7 Formal Bound Empirical Verification

Table 7 validates the bound from Eq. 1. At $\epsilon$=0.01, the theoretical flip probability is 0.076 while the empirical flip rate is 0.008, giving a tightness ratio of 9.5×. The bound becomes tighter at larger $\epsilon$ (3.1× at $\epsilon$=0.1), consistent with spectral-norm Lipschitz estimates being conservative for local perturbations. Table 8 shows the baseline comparison: without IMM, the theoretical bound exceeds 1.0 (vacuous), and the empirical flip rate is 66.5% at $\epsilon$=0.01—83× worse than IMM's 0.8%.

We stress that the bound's value is *prescriptive*: it correctly identifies $\Delta_k$ as the control variable for robustness, even though the absolute tightness is insufficient for runtime certification. This is consistent with known limitations of Lipschitz-based bounds in deep networks.

Table 6: Defense method comparison on LIBERO (average). IMM dominates the Pareto frontier of preservation vs. accuracy with minimal overhead.

| Method | Pres. $\epsilon$=0.01 | $\Delta$SR (pp) | Train Cost | Infer Cost | Target Selection? |
|---|---|---|---|---|---|
| No defense | 0.34 | 0 | 1× | 1× | No |
| Adv. Training | 0.52 | −18 | 3.2× | 1× | No |
| Rand. Smoothing | 0.61 | −22 | 1× | 8× | No |
| Jacobian Reg. | 0.48 | −12 | 2.1× | 1× | No |
| **IMM (ours)** | **0.88** | **−9.3** | **1.35×** | **1×** | **Yes** |

Table 7: Theoretical vs. empirical flip probability under IMM ($k$=72 of 144 tokens). The bound is directionally correct but 3–19× loose.

| $\epsilon$ | Theoretical $P$ | Empirical Rate | Tightness |
|---|---|---|---|
| 0.005 | 0.038 | 0.002 | 19.0× |
| 0.01 | 0.076 | 0.008 | 9.5× |
| 0.02 | 0.152 | 0.025 | 6.1× |
| 0.05 | 0.380 | 0.095 | 4.0× |
| 0.10 | 0.760 | 0.245 | 3.1× |

Table 8: Baseline vs. IMM flip rates. Without IMM, the theoretical bound is vacuous (>1.0). IMM reduces empirical flip rate by 83× at $\epsilon$=0.01.

| Targets | Base Theor. | Base Empir. | IMM Empir. | Improvement |
|---|---|---|---|---|
| 0.005 | >1.0 | 0.420 | 0.002 | 210× |
| 0.01 | >1.0 | 0.665 | 0.008 | 83× |
| 0.05 | >1.0 | 0.821 | 0.095 | 8.6× |

Table 9: Pruning ratio sweep on aloha_sim_insertion. The U-shaped G3 curve reveals a task-dependent optimal compression point.

| $k$ ratio | G3 | Preserv. | Action Loss | Speedup |
|---|---|---|---|---|
| 0.3 | 1.42 | 0.92 | 0.00578 | 3.3× |
| 0.4 | 1.08 | 0.91 | 0.00439 | 2.5× |
| 0.5 | 0.87 | 0.89 | 0.00354 | 2.0× |
| 0.6 | 0.82 | 0.87 | 0.00334 | 1.67× |
| 0.7 | 0.85 | 0.85 | 0.00346 | 1.43× |
| 0.8 | 0.91 | 0.82 | 0.00370 | 1.25× |
| 0.9 | 0.96 | 0.78 | 0.00391 | 1.11× |

## 4.8 Task-Adaptive Compression Ratio Selection

Table 9 presents a pruning ratio sweep on aloha_sim_insertion. The G3 curve exhibits a U-shape with a minimum at $k$=0.6 (G3=0.82), indicating a task-dependent sweet spot. At low $k$ (aggressive pruning), too many informative tokens are removed; at high $k$ (mild pruning), the denoising effect diminishes as more low-quality tokens are retained. Preservation monotonically increases with more aggressive pruning (0.78 at $k$=0.9 to 0.92 at $k$=0.3), creating an explicit tradeoff. On LIBERO, the U-curve shifts upward (all G3>1.0), and the optimal point becomes the *minimum degradation* ratio rather than an improvement point—motivating task-adaptive ratio selection.

## 4.9 Ablation Studies

**Training Mode Ablation.** Table 10 compares joint, gradient-isolated, and two-stage training on aloha_sim_insertion. Joint training (v7) achieves the highest G3 degradation (3.19×) due to backbone damage from margin loss gradients. Gradient isolation (v8) reduces this to 1.34× but implicit co-adaptation remains. Two-stage (v8b) achieves the lowest G3 (0.87×) with the highest margin ratio (93.2×). Preservation is similar across modes (0.89–0.92), confirming that robustness gains derive from the margin loss itself, while backbone quality depends critically on the training protocol.

**Annealing vs. Fixed Mask Ratio.** Table 11 compares curriculum annealing ($k$: 0.9 → 0.5) against fixed $k$=0.5. Annealing produces marginally higher G3 (0.91 vs. 0.87) and lower margin ratio (78.5× vs. 93.2×) with 3 additional hyperparameters. Preservation is equivalent (0.90 vs. 0.895). Conclusion: the added complexity is not justified.

**Token Importance Analysis.** Table 12 examines the distinction between pruned and retained tokens. The attention weight ratio is 0.69 (KS $p$=0.08, not significant at $\alpha$=0.05), meaning pruned tokens are not "noise" in the attention sense. However, the action gradient norm ratio is 0.43 ($p$=0.002, significant), indicating ACIS captures action-relevant information beyond attention weight. This supports the Pareto interpretation: ACIS identifies *moderately less relevant* tokens for removal, trading marginal action-relevance for substantial robustness gains—not "denoising."

## 4.10 Deployment Overhead

Table 13 quantifies ACIS deployment cost. The module adds 0.14% parameters and 2% latency, but 50% token pruning yields 27.4% net latency reduction and 28% memory savings. ACIS becomes a hard dependency: removing it at inference (bypass mode) degrades SR from 68% to 42%, because Stage 2 LoRA

Table 10: Training mode ablation on aloha_sim_insertion. Two-stage training is essential for preserving backbone quality.

| Mode | G3 | Margin× | Preserv. | Backbone |
|---|---|---|---|---|
| Joint (v7) | 3.19 | 64.3 | 0.912 | Severe damage |
| Grad-isolated (v8) | 1.34 | 50.2 | 0.923 | Implicit drift |
| Two-stage (v8b) | 0.87 | 93.2 | 0.895 | Preserved |

Table 11: Annealing ablation. Curriculum annealing adds hyperparameters without measurable benefit.

| Variant | G3 | Margin× | Preserv. | Extra HPs |
|---|---|---|---|---|
| Fixed $k$=0.5 (v8b) | 0.87 | 93.2 | 0.895 | 0 |
| Anneal 0.9→0.5 (v8c) | 0.91 | 78.5 | 0.900 | 3 |

adapts the backbone to expect ACIS-masked inputs. This architectural coupling is a limitation, though ACIS's negligible cost (2.8M params, 0.9ms) makes always-on deployment practical.

# 5 Discussion

**The Pareto Perspective.** IMM does not achieve lossless compression—it achieves *robustly degraded* compression. The 7–13pp SR drop is the explicit cost of transforming preservation from 33% to 87%. We argue this tradeoff is favorable for deployment: a system with 55% SR and 87% perturbation stability is more deployable than one with 65% SR and 34% preservation that can catastrophically fail under visual variation. The compression ratio $k$ and margin loss weight $\lambda$ serve as knobs for navigating this frontier.

**Limitations.** (1) The formal bound (Eq. 1) is 3–19× loose due to conservative Lipschitz estimation, limiting its use to a design criterion rather than a runtime certificate. (2) G3<1.0 is specific to aloha_sim_insertion; on LIBERO, G3 ranges 1.12–1.28, reflecting genuine accuracy cost. (3) ACIS is a hard dependency—bypass mode degrades SR by 36pp, requiring ACIS to be always-on at deployment. (4) Margin ratio improvements decrease with model scale (93.2× for 2B vs. 12.5× for 7B), suggesting scaling-aware margin objectives as future work. (5) Results are limited to simulated environments; real-world validation remains necessary.

Table 12: Token importance analysis. Pruned tokens have significantly lower action gradient norms but comparable attention weights, indicating learned importance captures action-relevance beyond attention.

| Metric | Pruned | Retained | Ratio | KS $p$ |
|---|---|---|---|---|
| Avg Attn Weight | 0.0031 | 0.0045 | 0.69 | 0.08 |
| Action Grad Norm | 0.0012 | 0.0028 | 0.43 | 0.002 |
| Position Entropy | 3.2 | 2.8 | 1.14 | 0.15 |

Table 13: ACIS deployment overhead. The module is lightweight and enables net efficiency gains through token pruning.

| Metric | Baseline | +ACIS | +ACIS+Prune | Net |
|---|---|---|---|---|
| Params (M) | 2000 | 2002.8 | 2002.8 | +0.14% |
| FLOPs/tok (M) | 850 | 851.2 | ∼425 | −50% |
| Latency (ms) | 45.2 | 46.1 | 32.8 | −27.4% |
| Memory (GB) | 8.2 | 8.3 | 5.9 | −28% |

# 6 Conclusion

We presented Importance Margin Maximization (IMM), a training framework that addresses the adversarial fragility of VLA token compression by directly maximizing the importance margin $\Delta_k$. IMM introduces a formal robustness bound that serves as a principled design criterion, a lightweight ACIS module (0.14% overhead), and a two-stage training protocol that prevents ACIS-backbone co-adaptation.

Our results demonstrate that IMM operates on a robustness-efficiency Pareto frontier: it trades a controlled 7–13pp success rate reduction for a +52–55pp improvement in adversarial preservation, transforming compression from a deployment-blocking vulnerability into a manageable tradeoff. IMM is compression-agnostic, generalizes across model scales (2B–7B), and dominates alternative defenses in both robustness and efficiency. The framework provides deployment engineers with explicit knobs—compression ratio and margin weight—to navigate task-specific tradeoffs.

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